

GSET Quantitative Execution Consultant — Prep Roadmap v2

Asia Equities Execution Solutions & Product · Prepared for Bill · Updated July 2026 after Round 1

What changed in v2 and why. Round-1 feedback: the role is **heavily statistics-focused**. This version reorganizes the roadmap around that signal: a statistics question bank with answer sketches (§3), the statistical probes your own CV invites (§4), a map of the statistics you can demonstrate live in your platform (§5), and a stats-sprint study plan (§6). The Round-1 context sections are retained in condensed form (§1–2); the original 10-topic ranking now reads: Topics 2 and 5 (applied statistics, experimental design) join Topic 1 (TCA) as the core — everything else supports them.

1. GSET Context (condensed from v1)

- **The business:** Goldman's electronic execution franchise — the algo suite (scheduling: VWAP/TWAP/POV; cost-minimizing: IS/Arrival, Almgren-Chriss-informed; liquidity-seeking: Sonar/Sonar Dark), the Smart Order Router with anti-gaming logic, the Sigma X dark-pool family (incl. Japan, Hong Kong, Australia), and the Atlas platform rebuild.
- **The seat:** Execution Consulting & Analytics — the metrics-driven bridge between clients (proof + customization), GSET R&D (spec proposals from analysis), and internal desks. Your triangle: statistical fluency, microstructure intuition, plain-English delivery.
- **The rhythm:** Asia hours; morning TCA outlier scan → client/desk calls → midday statistical work (tests, regressions, time series) → afternoon backtests and A/B design → documentation for reproducibility and audit.

2. What 'Good' Looks Like (unchanged — now with a statistics reading)

- A quantitative conclusion in under two minutes — which, statistically, means you lead with the effect size and n , and the caveats are structural (design), not verbal hedging.
- Noise-vs-signal instinct = a power calculation you can do in your head (see §3 Q2 — memorize the worked example).
- Moving between code and slide without losing rigor = every chart carries its n , its standard error convention, and its selection story.

3. The Statistics Question Bank (the round-2 core)

Each answer sketch is written to be spoken in 60–90 seconds. Practice them aloud; the design point always comes before the test statistic.

Q: How would you test whether Algo A beats Algo B on slippage?

Answer sketch: Design before test: pair the comparison — same days, same order profiles — so day effects and regime cancel; then a paired t-test on the differences, with Wilcoxon signed-rank alongside because cost distributions are fat-tailed and right-skewed. Report the effect in bps with a confidence interval, not just a p-value. On real client flow the two algos never ran on identical orders, so the honest version is a regression: cost on a strategy dummy plus controls — size, volatility, participation, spread — where the dummy's coefficient is the incremental cost holding conditions fixed. I'd show the naive difference next to the controlled one so the confounding adjustment is visible.

Q: How many orders do you need to detect a 2 bps improvement?

Answer sketch: Paired design: $n = ((z_{\alpha/2} + z_{\beta}) \cdot \sigma_d / \delta)^2$. With σ of paired slippage differences ≈ 30 bps, $\delta = 2$ bps, 5% size, 80% power: $n = ((1.96 + 0.84) \cdot 30 / 2)^2 \approx 1,760$ paired orders. That number is the point: at realistic dispersion, small edges need thousands of orders — which is why algo wheels accumulate quarters of data before re-ranking, why 'not statistically separable at this n' is often the correct readout, and why anyone claiming a 2 bps win off 50 orders is selling noise.

Q: Why robust standard errors — and which kind, when?

Answer sketch: Execution-cost residuals violate homoskedasticity by construction — variance scales with order size and volatility — so White/HC1 at minimum. Costs measured over time carry serial correlation (regimes persist), so Newey-West HAC when the panel has a time dimension. And when many orders share the same day, they share the day's shock — cross-sectional correlation — so cluster by day (or by event date in event studies). The interview trap is naming HC1 and stopping; clustering is usually the binding issue on desk data.

Q: You run 8 algos — how do you handle multiple comparisons?

Answer sketch: Three layers. Within one comparison across 8 algos, that's 28 pairwise tests — control the family-wise error with a rank-based omnibus first (Friedman), then a Nemenyi-type critical difference on average ranks rather than raw pairwise t-tests. Across many studies — re-running the wheel for many clients or configs — control the false discovery rate with Benjamini-Hochberg on the study-level p-values. And sequentially: don't peek — fixed horizons or alpha-spending, because stopping when a test 'goes significant' inflates size dramatically.

Q: Cost data is nowhere near normal. What actually breaks, and what do you do?

Answer sketch: Means are dominated by the tail — one limit-up day distorts a month — so report medians and trimmed means alongside; use rank tests (Wilcoxon, Friedman) which trade a little power for a lot of robustness; bootstrap the confidence intervals rather than relying on t-approximations; and for tail-risk questions use quantile regression — clients care about the 95th percentile cost at least as much as the mean. Normality of residuals (Jarque-Bera) matters for small-sample inference, not for coefficient consistency — know which claim you're making.

Q: A client's monthly TCA is 8 bps worse than last month. Is the algo worse?

Answer sketch: Decompose before concluding. First mix: did order size, side, market, or urgency composition shift? An 8 bps swing is routinely a mix effect. Second regime: volatility and spread levels. Third, the statistical question: given this month's n and dispersion, is 8 bps outside noise at all? The clean tool is an expected-cost benchmark — predict each order's cost from a fitted model given its conditions, then compare realized vs predicted month over month. If realized-minus-predicted deteriorated, now you have an algo conversation; if predicted moved, you have a flow conversation.

Q: How do you design an A/B test for an algo change?

Answer sketch: Randomize at the order level within client-market-size strata, so both arms face the same regime mix; pre-register the metric (IS vs arrival, say), the horizon, and the sample size from a power calculation; block by day. Guard the classic trading confounds: time-of-day, venue mix, and volatility clustering. Analyze as designed — paired/blocked — and report both the effect and what you could NOT detect at this n. If randomization isn't possible (client won't split flow), fall back to regression with controls and say plainly that it's observational.

Q: How do you attach significance to an event study — one stock, one event?

Answer sketch: Single-firm Brown-Warner: the estimation window's residual variance gives the null distribution for event-window abnormal returns, with the Patell-style correction for forecast error — variance inflates by $(1 + 1/L + \text{the market-return distance term})$. CAR variance is the running sum, assuming serially uncorrelated residuals — the market model's own assumption. Two honest caveats: event-induced variance makes these bands anti-conservative on the event days

themselves (the BMP critique), and with clustered events — many names, same rebalance date — the cross-section is correlated, so you cannot just average t-stats.

Q: What time-series statistics do you actually use on execution data?

Answer sketch: Stationarity checks before regressing anything trended; autocorrelation structure — Ljung-Box on returns, and on cost residuals to justify HAC lags; and for regime work, the Lo-MacKinlay variance-ratio test with the heteroskedasticity-robust z^* — it formalizes 'is this tape trending or mean-reverting' better than any moving-average heuristic, and it's what I use for trend classification in my own platform.

Q: Give me a selection-bias story from execution analytics.

Answer sketch: Three I've actually handled. Endogenous strategy choice: traders route hard orders to IS in stressed tape, so IS 'looks' expensive raw — that's the flow, not the engine; controls or randomization fix it. Conditioning on the outcome: in my Taiwan limit-up study, sampling stocks that hit the limit and asking what happened next builds survivorship into the panel — the fix is defining the at-risk set ex ante. Look-ahead: volume curves built including the simulated day leak the answer into the schedule — mine are built strictly from prior days.

4. The Statistical Probes YOUR CV Invites (prepare these cold)

Q: Your Invesco study found +2% T+1 after limit-up retreats. Was that significant?

Answer sketch: The honest answer, delivered confidently: as run, it was a raw mean over a small, event-clustered sample — no market adjustment, no clustered errors, so I would not defend it as a tested effect. What I'd do now, and have built since: market-model abnormal returns instead of raw; bootstrap or date-clustered errors because all names on one rebalance date share the day's shock; split locked vs retreat regimes; and state n per cell. The pattern was strong and intuitive enough to act on operationally, but the inference discipline came later — and that gap is literally why my current platform puts confidence bands on every event-study number.

Q: Your intraday thresholds — how were they validated?

Answer sketch: In-sample, by scan — which risks overfitting, and I flagged the false-positive problem at the time. The right formalization is a hazard model: $P(\text{hits limit} \mid \text{return-so-far, time-of-day, volume-so-far})$, fit walk-forward with by-year splits, evaluated on precision-recall with an explicit cost on false positives — accelerating into a name that doesn't lock is a real cost, not a free option. That's the named next build on my platform's roadmap.

Q: Only ~140 days of intraday data — what does that limit?

Answer sketch: Bloomberg's retention cap, and it binds exactly where power matters: intraday threshold estimation across enough independent rebalance events. Mitigations: the daily-data analysis extends back a decade (2015/2016) so the T+1 economics rest on more events; and the durable fix is accumulation — my platform's event library records every studied event so the priors sharpen each cycle instead of resetting.

Q: Isn't your platform's data too clean to say anything about real fills?

Answer sketch: The panel is simulated, and I say so on every page — but the statistical machinery is chosen precisely so it transfers: the cost regression, robust and clustered errors, the blocked wheel design, the power arithmetic — all of it runs unchanged on a real fill panel. Simulation is what lets me pin every method with deterministic tests — 181 of them — so when it meets real data, the code is the already-validated part.

5. The Statistics You Can Demo Live (your artifact advantage)

Statistical method	Where it lives in the platform	One-line demo script
OLS + HC1/Newey-West, F, DW/BP/JB	Cost Model — TCA Regression (Page 1)	"Fitted, not assumed — every SE auditable."
A/B with controls (dummy + confounders)	Cost Model section	"Naive vs controlled shown together."
Condition-adjusted ranking	Cost Model — wheel-defense view	"Raw rank last, adjusted rank first — same data."
Friedman + Nemenyi CD (blocked design)	Algo Wheel league table	"Not separable at this n IS the finding."
Paired t / Wilcoxon / bootstrap	Hypothesis Testing (Agent 10)	"Design first: same days, both arms."
Brown-Warner event inference	Page 2 CAR bands + t column	"Bands, not points — and anti-conservative where."
Variance-ratio (Lo-MacKinlay) trend test	Agent 2 regime	"A test, not a moving average."
Ljung-Box, ACF, Amihud, EDGE/Roll/CS/AR	Microstructure analytics	"Four spread estimators; disagreement is information."
Power / sample-size discipline	Wheel notes + this doc §3 Q2	"~1,760 pairs for 2 bps at $\sigma=30$."

Demo order for a statistics-focused interviewer: condition-adjusted ranking → wheel CD → IS attribution reconciliation → CAR bands. Fifteen minutes, all inference.

6. Revised Study Plan — the Statistics Sprint (7 days)

Day	Focus	Self-check
1–2	§3 Q1–Q5 aloud until fluent; re-derive the power formula and the worked 1,760 example on paper	Cold: design + test + effect size for "A vs B" in 90 seconds
3	Regression depth: robust vs clustered SEs, diagnostics, what each violation breaks	Explain when HC1 is NOT enough and clustering is
4	Multiplicity + sequential testing; Friedman/Nemenyi mechanics; BH	Compute a Nemenyi CD for $k=8$, $n=60$ from the formula
5	Event-study inference + your Invesco self-critique (§4) verbatim	Deliver the +2% answer without defensiveness
6	Platform demo run-through (§5 order); rehearse one-line demo scripts	15-minute demo lands all four inference moments
7	Mock: mixed stats + microstructure + the client-case (§3 Q6); day-before checklist	All checklist derivations from memory

7. Day-Before Checklist (v2)

- Re-derive from memory: implementation shortfall (Perold components), the paired-test power formula with the 1,760 worked example, the Nemenyi CD formula, and the Brown-Warner variance with its correction term.

- Say aloud: the robust-vs-clustered SE distinction; the three selection-bias stories; the Invesco self-critique.
- Pre-warm the platform demo (cache the ticker; seed the event library ≥ 3 events; VEDL study run once).
- Questions to ask her, statistics-flavored: how the team sets sample-size discipline on wheel readouts before re-ranking brokers; how they control false discovery when running many client studies in parallel; whether Atlas changes what's logged for analytics (order-state granularity is the binding constraint on good TCA).